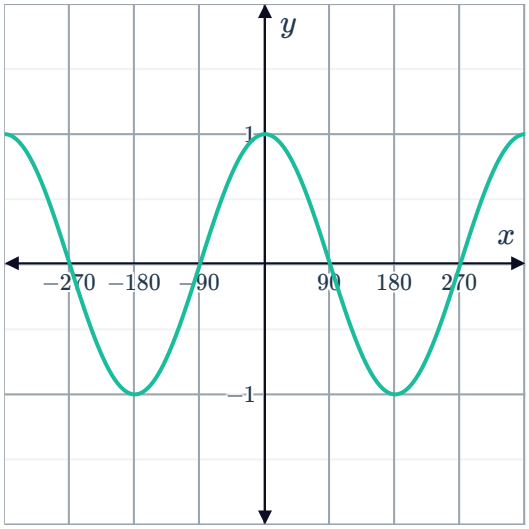
Vertical dilations

1

a

θ	0	60	90	120	180	240	270	300	360
$\cos\theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e $[-4, 4]$

2

a $(-\infty, \infty)$

b $[-1, 1]$

3

a $(-\infty, \infty)$

b $[-2, 2]$

4 8

5

a $y = 2\sin x$

b $y = -\sin x$

c $y = 3\cos x$

d $y = -1.5\cos x$

e $y = 3\sin x$

f $y = 2.5\sin x$

6 270

7

a 8

b 0

c -8

8 5

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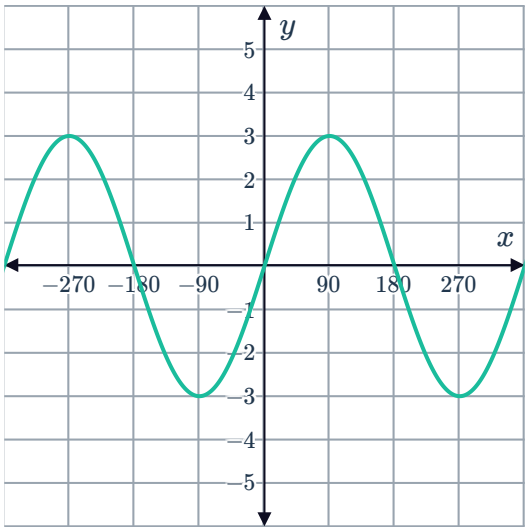
9 $-2\sin x$

10

a

i $A = 3$

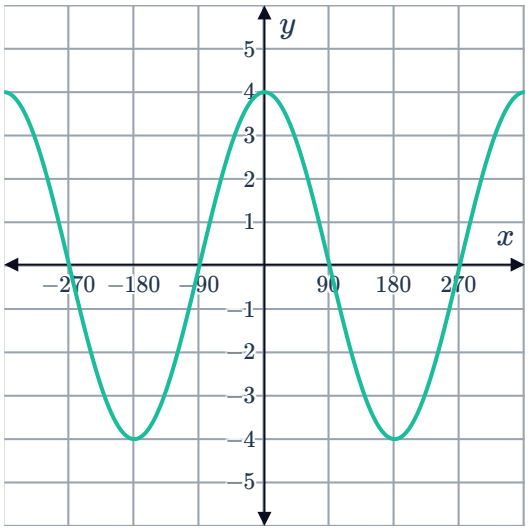
ii



b

i $A = 4$

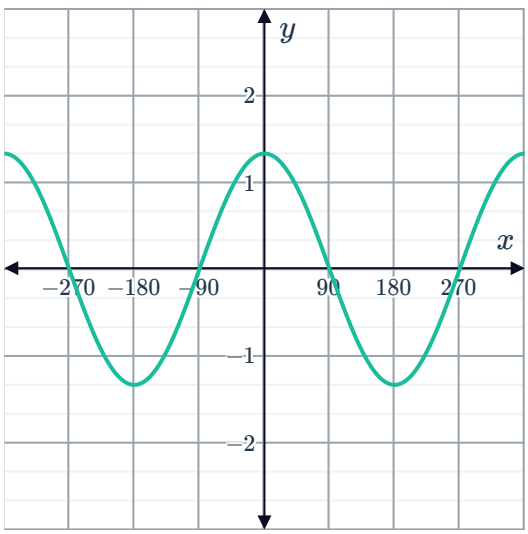
ii



c

i $A = \frac{4}{3}$

ii

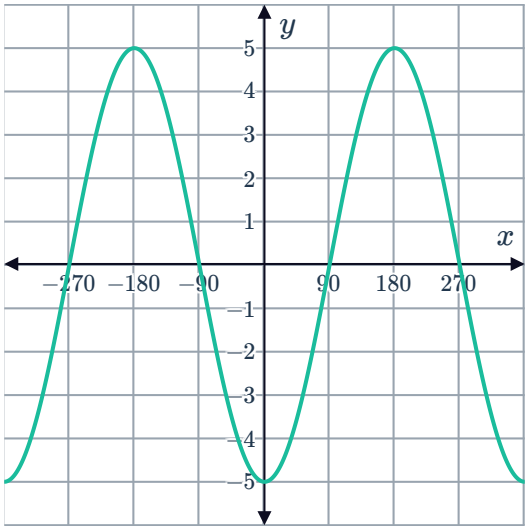


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d

i $A = 5$

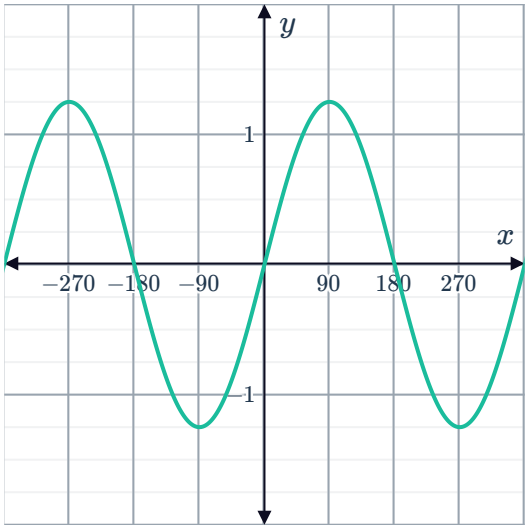
ii



e

i $\frac{5}{4}$

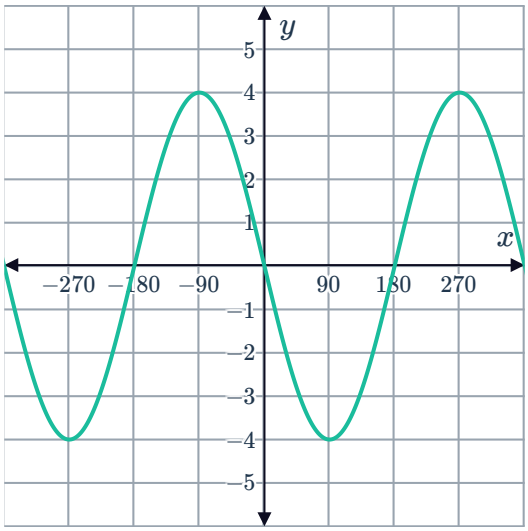
ii



f

i $A = 4$

ii



11

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b -3

c 3

d A vertival dilation by a factor of 3 and reflection about the x-axis.

Vertical translations

12 Translated 2 units downwards.

13 Translated 2 units upwards.

14 Translated 4 units upwards

15

a $y = \sin x + 4$

b $y = \sin x - 2$

c $y = \sin x - 4$

d $y = \sin x + 1$

16

a $y = \cos x + 3$

b $y = \cos x - 4$

c $y = \cos x - 3$

d $y = \cos x + 2$

17 Reflection about the x-axis, then translation 3 units up.

18 $y = 2$

19

a Translated 4 units upwards.

b 360°

c 5

d 3

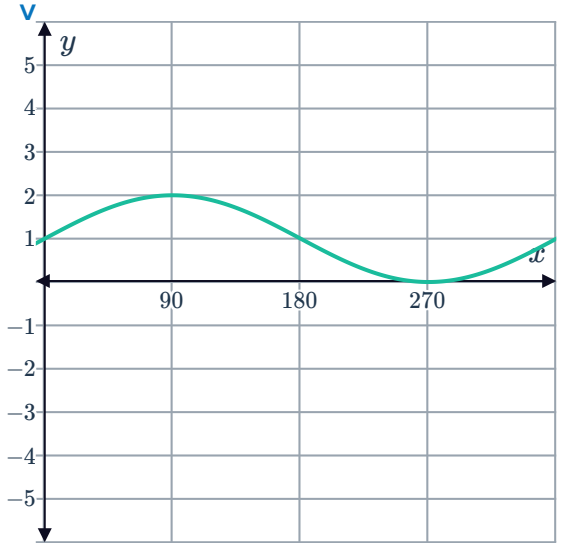
20

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a

i 360 ii 1 iii 2 iv 0

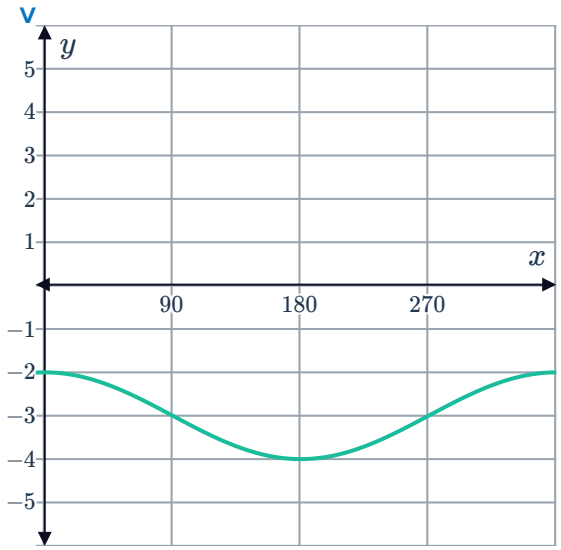
v



b

i 360 ii 1 iii -2 iv -4

v

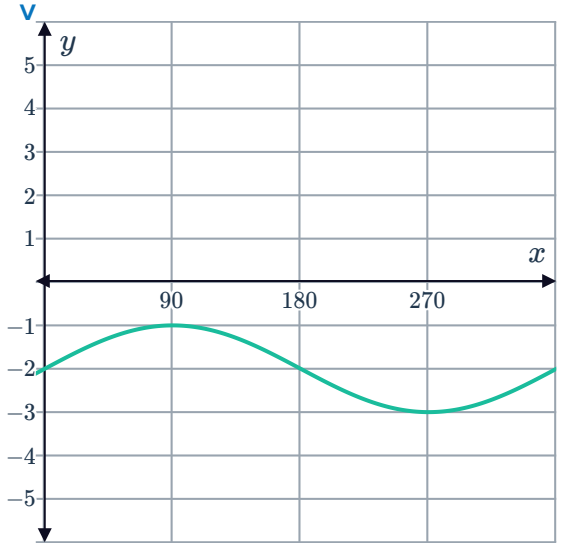


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c

i 360 ii 1 iii -1 iv -3

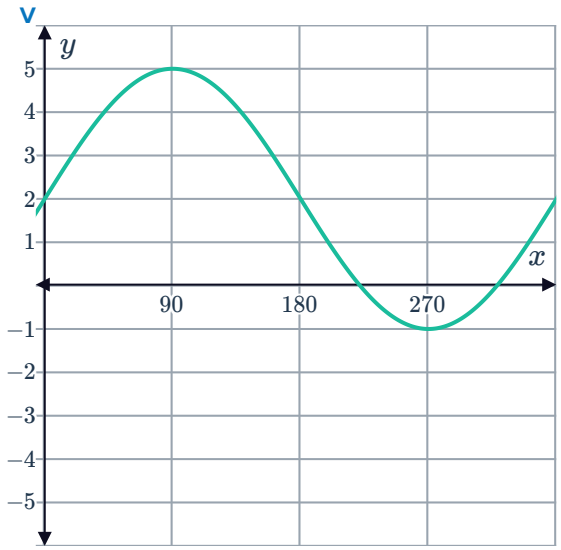
v



d

i 360 ii 3 iii 5 iv -1

v

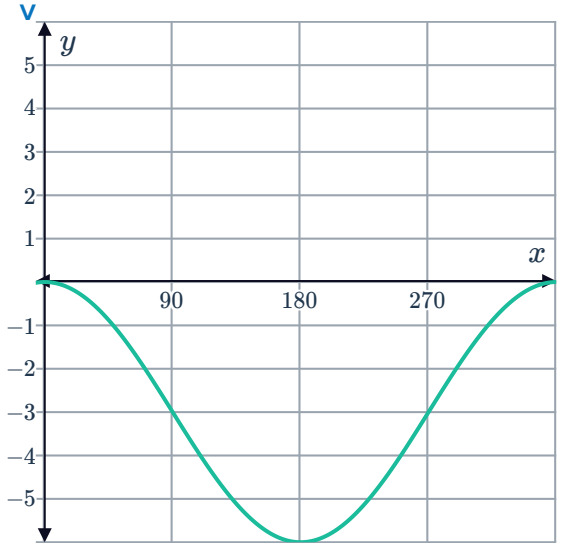


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e

i 360 ii 3 iii 0 iv -6

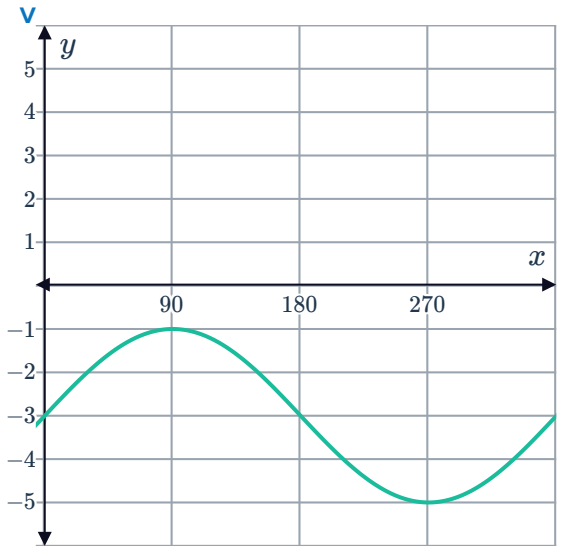
v



f

i 360 ii 2 iii -1 iv -5

v



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21

a

i 3

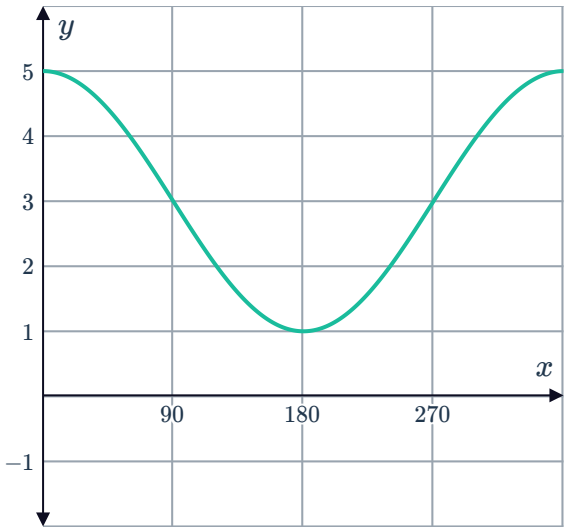
ii 2

iii 5

iv 1

v A vertical dilation by a factor of 2, then translated 3 units upwards.

vi



b

i -3

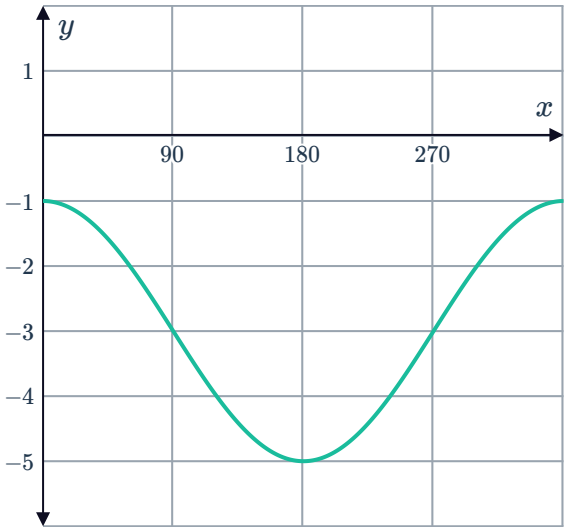
ii -1

iii -5

iv 2

v A vertical dilation by a factor of 2, then translated 3 units downwards.

vi



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c

i 2

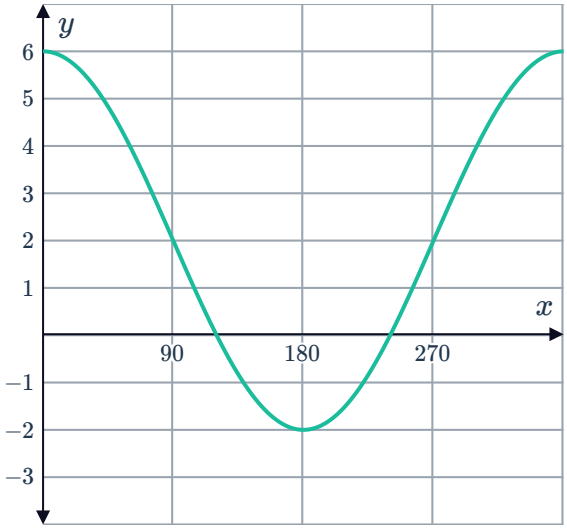
ii 6

iii -2

iv 4

v A vertical dilation by a factor of 4, a reflection about the x-axis and then translated 2 units upwards.

vi



d

i 2

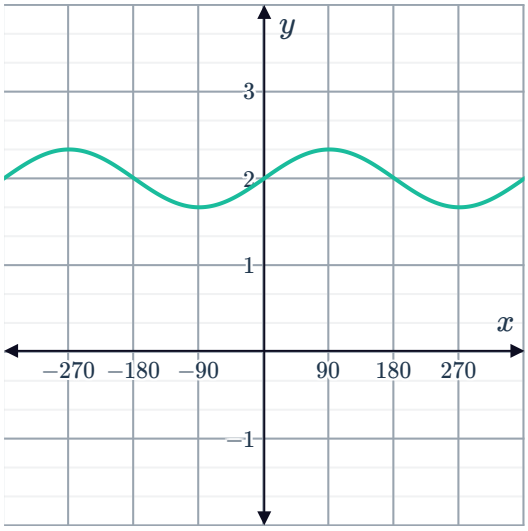
ii $\frac{7}{3}$

iii $\frac{5}{3}$

iv $\frac{1}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

vi



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22 **a** $y = \cos x + 9$ **b** 10

23

a **i** 5 **ii** -1 **b** **i** 7 **ii** 1 **c** **i** 2 **ii** 1 **d** **i** 2 **ii** $\frac{1}{2}$

24 $2\sin x + 2$

25 $8\cos x - 2$

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